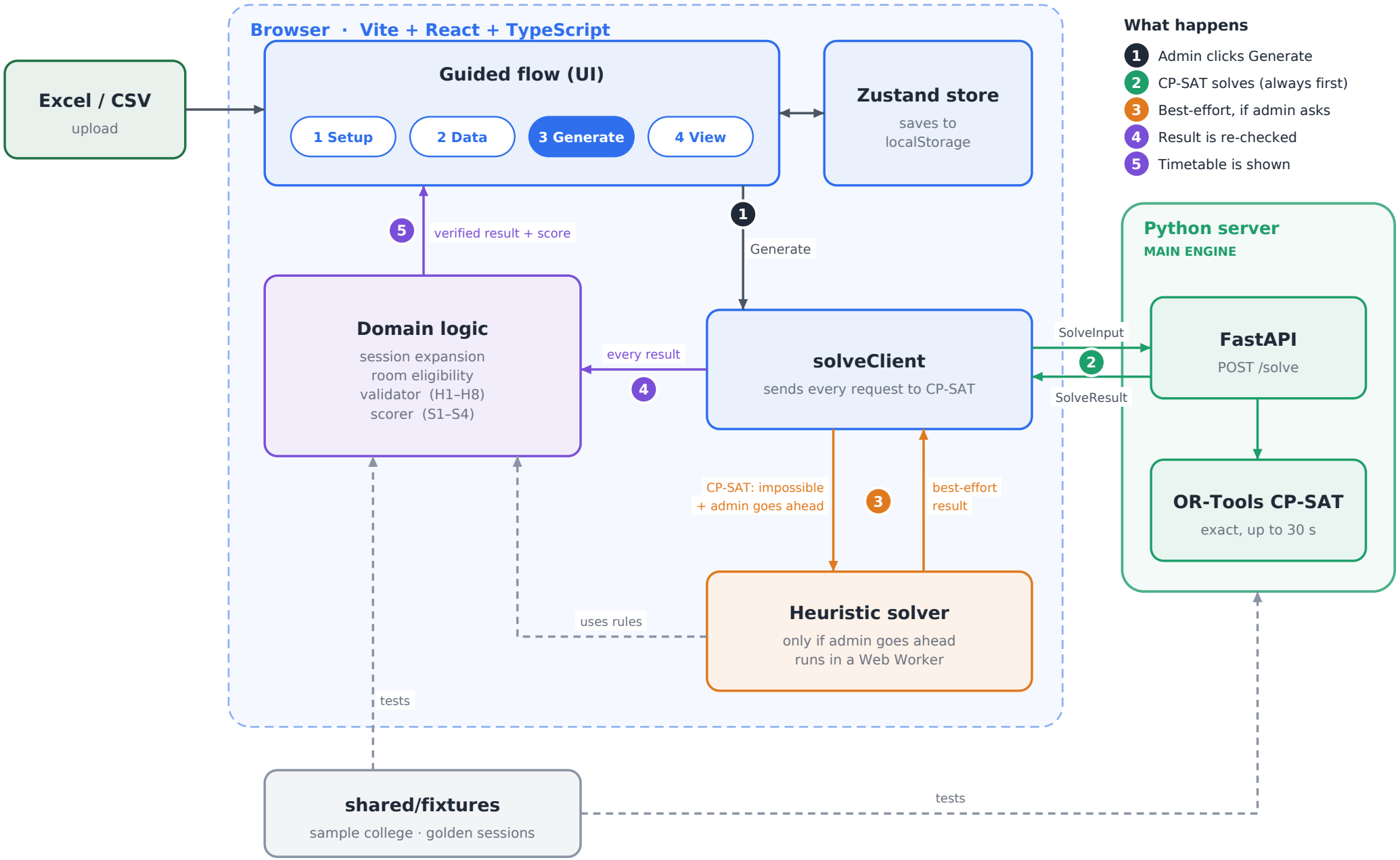
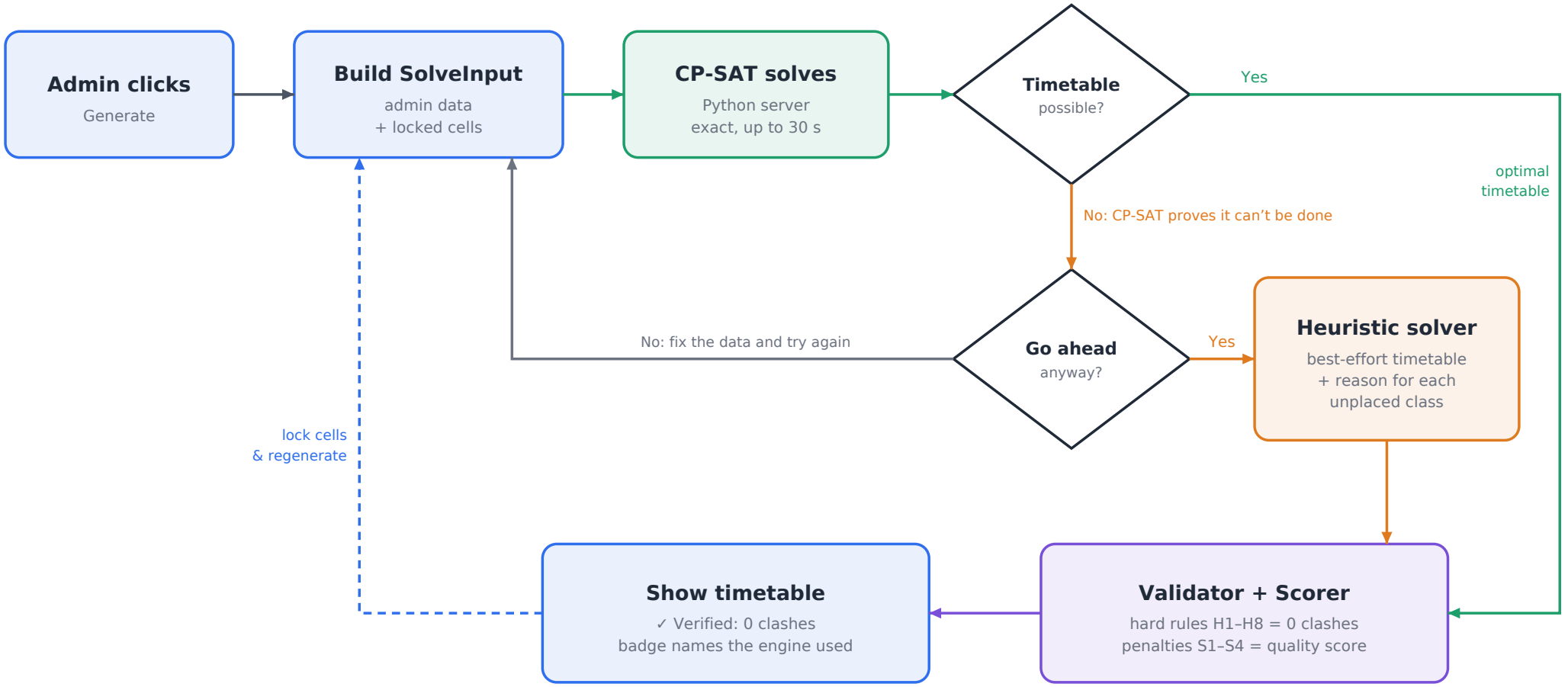
System architecture

CP-SAT on the Python server is the main engine. If it finds no valid timetable, the admin can ask the browser heuristic for a best-effort one.



How a timetable is generated

CP-SAT always runs first. The heuristic runs only when CP-SAT finds no valid timetable and the admin chooses to go ahead anyway.



Hard rules: never broken

H1–H3	no teacher, section or room double-booked
H4	room matches session type and department
H5	every session placed exactly once
H6–H8	labs in consecutive periods · nothing in breaks · daily lunch

Soft rules: penalised in the score

S1	same subject twice in a day
S2	teacher teaches more than 3 periods in a row
S3	idle periods in a section’s day
S4	a batch sits idle while another is in a lab